

Keyboard-to-Pico wire list

Bread80 CPC464 SMT / J3 19-pin connector / original RP2040 Pico or Pico H

Keyboard disconnected from CPC. Use 19 signal wires only. Do not connect a matrix contact to power or ground. Power the Pico through its USB connector.



J3 pad	Matrix net	Pico GPIO	Pico header pin	Signal role
01	X9	GP20	26	Sense input
02	Y10	GP11	15	Scan output
03	X1	GP12	16	Sense input
04	X2	GP13	17	Sense input
05	X3	GP14	19	Sense input
06	X4	GP15	20	Sense input
07	X5	GP16	21	Sense input
08	X6	GP17	22	Sense input
09	X7	GP18	24	Sense input
10	X8	GP19	25	Sense input
11	Y9	GP10	14	Scan output
12	Y8	GP9	12	Scan output
13	Y7	GP8	11	Scan output
14	Y6	GP7	10	Scan output
15	Y5	GP6	09	Scan output
16	Y4	GP5	07	Scan output
17	Y3	GP4	06	Scan output
18	Y2	GP3	05	Scan output
19	Y1	GP2	04	Scan output

Header pin is the physical pin number. For example, J3 pad 1 connects to **GP20 on Pico pin 26**, not to Pico pin 20.

Identify J3 before wiring

With keys facing you and cursor keys at upper right, J3 is the single vertical row at the keyboard's far left. On the inspected PCB, **pad 1 is square at the Esc end**; pad 19 is at the Caps end. The solder-side view is mirrored.

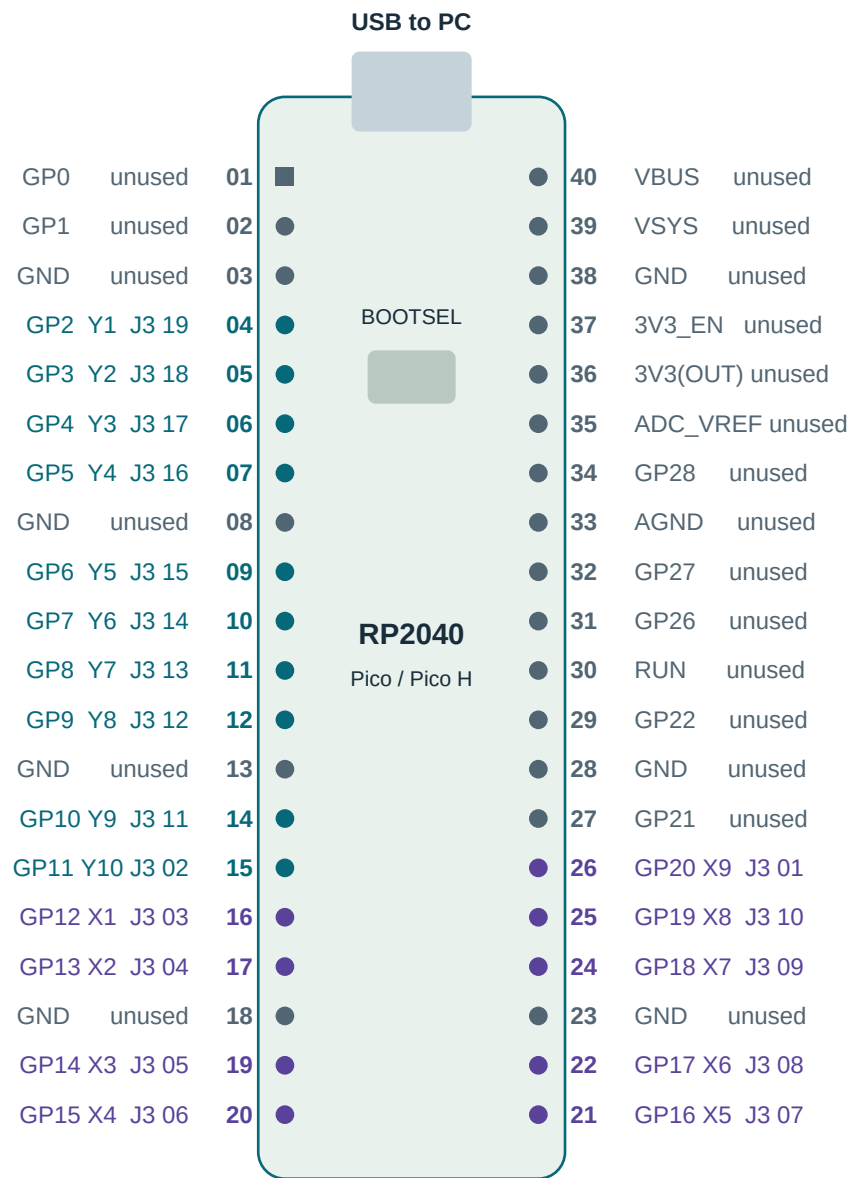
Leave **LK1, LK2, LK3 and LK4-LK6 open** for this prototype. LK3 joins X8/X9; the last three links serve the unused bonus keys. **J1 and J2 have different pin orders.**

J3 nets: Bread80 PCB revision bf4a6052888f. Firmware pin assignments: src/board_config.h. See page 4 for rgbwalker V1.2 checks and the first hardware test.

Pico header orientation

Top view: components and BOOTSEL button facing you, USB connector at the top. Not to scale.

Numbers beside the board are **physical header pins**. Labels show GPIO / matrix net / J3 pad. Anything marked **unused** stays unconnected in this keyboard harness.



Y = scan outputs **X = sense inputs** Grey = unused

Three details that prevent wiring mistakes

Count down the left side: 1 to 20. The right side counts up from 21 at the bottom to 40 at the top. Ground pins interrupt the GPIO sequence.

No matrix supply pin: X inputs use the Pico's internal 3.3 V pull-ups. The USB cable powers the Pico; the passive keyboard needs only the 19 signal connections.

Do not join X8 and X9 in this harness. X9 is dedicated to DEL. Both Shift switches share Y3/X6 and cannot be distinguished by firmware.

Physical numbering checked against [Raspberry Pi's official Pico pinout PDF](#). The diagram above is an original drawing for this project.

Assemble and verify

Work with USB unplugged during wiring and meter checks. Connect only the passive keyboard.

- 1. Prepare:** identify the exact board and J3 pad 1. Open LK1-LK6. Fit Pico headers if needed. Use insulated wires or a labeled breakout harness; keep the first cable short.
- 2. Wire:** follow page 1 one contact at a time. Check every cable endpoint and look for unintended adjacent shorts. Do not rely on ribbon-wire colors or a mirrored photograph.
- 3. Test contacts:** disconnect the keyboard from both Pico and CPC; check the pairs below. On a diode board, place the red meter probe on X and black on Y, using diode mode.

Key held	Y / X nets	J3 pads	Pico header pins
Cursor Up	Y1 / X1	19 / 3	4 / 16
A	Y9 / X6	11 / 8	14 / 22
Space	Y6 / X8	14 / 10	10 / 25
Either Shift	Y3 / X6	17 / 8	6 / 22
DEL	Y10 / X9	2 / 1	15 / 26
Main Enter	Y3 / X3	17 / 5	6 / 19
Keypad Enter	Y1 / X7	19 / 9	4 / 24

A pressed key should conduct; releasing it should open the path. A diode blocks the reverse direction. A plain switch conducts either way. A continuity beeper may not detect the diode path reliably.

Current path: internal 3.3 V pull-up → X → closed switch → diode → selected Y.
Diode cathode faces Y. Only the selected Y is driven low; inactive Y pins float.

Choose the correct scan option

Bread80 SMT, all diodes verified: set the option ON. **rgbwalker, membrane, tactile, or uncertain diodes:** keep it OFF. These commands assume the SDK build is already configured; see README.md for initial setup.

```
cmake -S . -B build -DGOODJOHN_MATRIX_HAS_DIODES=ON
cmake --build build --parallel
```

Use **OFF** instead of ON for a matrix without verified diodes.

Flash, then test on the PC

Hold BOOTSEL while connecting the Pico, then copy **build/goodjohn.uf2** to RPI-RP2. Use a USB data cable. Check every key and release, modifiers, DEL and both Enter keys. COPY sends Alt; DEL sends Backspace; CLR sends forward Delete. Set the PC's Num Lock for keypad digits.

If one group of keys fails, recheck its X/Y wire and diode orientation. If only chords fail on a board without diodes, ghost suppression may be blocking ambiguity. More than six regular keys produces HID ErrorRollOver. Shifted symbols follow the PC layout; there is no Fn layer yet.

Full keyboard, cable settling, suspend current and suspend/resume still need bench tests. One directly wired key on rgbwalker V1.2 has typed correctly over USB; see page 4. No USB remote wakeup or low-power suspend implementation is provided yet.

Other boards and source notes

The logical CPC key matrix is shared; connector order and jumper routing are not universal.

rgbwalker V1.2: first key test passed. On 2026-10-06, top-row **1** typed **1** with two direct wires. The full keyboard remains unverified.

V1.2 Gerbers connect CP002_THT and CP002_SMD contacts 1-19 in the same order; contact 20 is isolated from the matrix. Count the **square pad as 1 at the Esc end**, toward Shift. Check other revisions separately.

Expected contact(s)	Expected switch lines	Pico harness, if confirmed
1 and 2	Isolated DEL switch	GP20 and GP11
3 through 10	X1 through X8	GP12 through GP19
11 through 19	Y9 through Y1	GP10 down through GP2
20	No matrix connection (V1.2)	Leave isolated

Start directly: keyboard **3 to Pico physical 16 (GP12)**; keyboard **11 to physical 14 (GP10)**. With USB unplugged, each wire must beep pad-to-pad without a key held. Connect USB: top-row 1 should type 1. Bypassing the earlier ribbon/jumper assembly restored this; its exact fault is unresolved.

Unplug USB, then add **4 to physical 17 (GP13)** for 2; next **5 to physical 19 (GP14)** for Esc. Check each wire before power. Esc prints no character. Complete contacts 1-19 using page 1, leave 20 disconnected, and keep **GOODJOHN_MATRIX_HAS_DIODES=OFF**. Details: docs/wiring.md.

Bread80 alternatives

J1 membrane / J2 modular: different pad orders, with routing affected by links. This sheet does not map those connectors. **Tactile board:** same X/Y scheme but different connector references and no diodes. **CPC6128 SMT:** upstream marks it as a design-stage board; it is not a validated adapter target.

An 18-wire adaptation must handle the shared X8/X9 net explicitly; simply omitting X9 loses DEL.

Source records

Bread80 CPC464 SMT KiCad PCB

Revision bf4a6052888fef27ca01a6b0983791d7cb4e8af1. J3 pad nets, square pad 1, connector placement, diode cathodes and switch coordinates were inspected.

rgbwalker V1.2 fabrication archive

Revision 588e61006bb606d08cc981467d7990c6e3104c42. Copper/plated-hole tracing checks THT/SMD correspondence, unused contact 20, and switch pairs: 1 = 3/11, 2 = 4/11, Esc = 5/11.

Raspberry Pi official Pico pinout

Pico-R3-A4-Pinout.pdf. Used to check the 40 physical header positions and power/ground labels. Target is the original non-wireless RP2040 Pico/Pico H.

Files and regeneration

The companion ASCII sheet is **docs/pinout.txt**. The generator reads the firmware GPIO arrays in **src/board_config.h** and produces both formats from one wiring dataset. These are original vector drawings, not scaled PCB templates.

```
python3 -m pip install -r scripts/requirements-docs.txt
python3 scripts/generate_pinout.py
```